Structured notes: Module IV: dielectric and magnetic properties of solids

Puri and Babbar, *Solid State Physics and Electronics*, chapters 8 and 9 (printed pages 228–279). Text extracted by Codex from the page images and checked against them. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Chapter questions and problems are not included.

Section 1: magnetic terminology and types of magnetism (8.1–8.2)

8.1–8.2 Magnetic terminology and types of magnetism (structured notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 228–230. Equations 8.1 to 8.8 are cited by their book numbers.

Before you start

- Magnetic moment of a current loop, $m = IA$.
- Magnetic field $\mathbf{B}$ and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

> **PLUS TWO RECALL: MAGNETIC MOMENT AND MAGNETIC MATERIALS**
>
> **Recall.** A current loop of area A carrying current I has moment $m = IA$. Materials are classed as diamagnetic, paramagnetic or ferromagnetic by their response to a field.
>
> **Re-derive.**
>
> 1. The magnetisation $\mathbf{M}$ is the moment per unit volume.
> 2. For weak fields $M \propto H$, so $\chi = M/H$ is a constant of the material.
> 3. Negative and small χ (diamagnetic), small positive χ (paramagnetic), and large positive χ (ferromagnetic) are the classes.
>
> **Saved.** Eq. (8.1).

8.1 Magnetic terminology

The magnetisation $\mathbf{M}$ is the magnetic moment per unit volume. The susceptibility is

$$\chi = M/H \tag{8.1}$$

χ is a tensor in general and a scalar in isotropic media. It is dimensionless in SI units. The molar susceptibility χ_m refers to a gram molecule.

The flux density $\mathbf{B}$ in the medium is

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) \tag{8.2}$$

$$\mathbf{B} = \mu_0(1 + \chi)\mathbf{H} \tag{8.3}$$

$$\mathbf{B} = \mu\mathbf{H} \tag{8.4}$$

$$\mu = \mu_0\mu_r \tag{8.5}$$

$$\mathbf{B} = \mu_0\mu_r\mathbf{H} \tag{8.6}$$

$$\mu_r = 1 + \chi \tag{8.7}$$

In free space $\chi = 0$, $\mu_r = 1$ and $\mathbf{B} = \mu_0\mathbf{H}$ (8.8).

$$\mathbf{B} = \mu_0\mathbf{H} \tag{8.8}$$

8.2 Types of magnetism

Magnetism in solids arises from the orbital and spin motion of electrons, and to a smaller extent from nuclear spins. The main contribution is the spin of unpaired valence electrons, which gives permanent electronic moments. The total moment depends on the number of unpaired electrons in each atom and on the relative orientation of neighbouring moments. Five classes are recognised: diamagnetism (induced, opposing the field, very weak), paramagnetism (permanent moments, aligned with the field, weak), ferromagnetism (parallel alignment of neighbours, very strong), antiferromagnetism (antiparallel and equal, so no net moment) and ferrimagnetism (antiparallel and unequal, so a net moment). Above the critical temperature all materials are paramagnetic.

Section 2: diamagnetism, classical and quantum (8.3, Example 8.1)

8.3 Diamagnetism (structured notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 230–235 (§8.3, classical and quantum theories; Example 8.1). Equations 8.9 to 8.28 are cited by their book numbers. The quantum theory (§8.3.2) is background to item 18 and is kept in full.

Before you start

- Lenz's law (Plus Two recall below).
- Circular orbit dynamics: Coulomb attraction supplies the centripetal force.
- Magnetic moment of a current loop; the vector potential (background).

> **PLUS TWO RECALL: LENZ'S LAW**
>
> **Recall.** An induced current opposes the change in flux that produces it.
>
> **Re-derive.**
>
> 1. An orbiting electron is a current loop with a moment.
> 2. A switched-on field changes the flux through the loop.
> 3. The induced moment opposes the field, so χ is negative (Eq. 8.16).
>
> **Saved.** Eqs. (8.16) and (8.20).

8.3.1 Langevin's classical theory

For an electron of radius ρ about a nucleus of charge Ze:

$$m\omega_0^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2} \tag{8.9}$$

$$\omega_0 = \sqrt{\frac{Ze^2}{4\pi\varepsilon_0 m\rho^3}} \tag{8.10}$$

A field $\mathbf{B}$ perpendicular to the orbit exerts a Lorentz force

$$\mathbf{F} = e(\mathbf{v}\times\mathbf{B}) \tag{8.11}$$

with magnitude

$$F = e\rho\omega B \tag{8.12}$$

The force balance with the field is

$$\begin{aligned} m\omega^2\rho &= \frac{Ze^2}{4\pi\varepsilon_0\rho^2} - e\rho\omega B \\ &= m\omega_0^2\rho - e\rho\omega B \end{aligned} \tag{8.13}$$

For $eB/2m \ll \omega_0$ the solution is

$$\omega \cong \pm\omega_0 - \frac{eB}{2m} \tag{8.14}$$

This is Larmor's theorem: the frequency changes by $eB/2m$. The additional current and moment are

$$\begin{aligned} I &= \text{charge} \times \text{revolutions per unit time} \\ &= (-e)\left(\frac{1}{2\pi}\frac{eB}{2m}\right) \\ &= -\frac{e^2B}{4\pi m} \end{aligned} \tag{8.15}$$

$$\mu_e = -\frac{e^2B}{4m}\rho^2 \tag{8.16}$$

$$\mu_a = -\frac{Ze^2B}{4m}\langle\rho^2\rangle \tag{8.17}$$

$$\langle r^2\rangle = \frac{3}{2}\langle\rho^2\rangle \tag{8.18}$$

Hence

$$\begin{aligned} \mu_a &= -\frac{Ze^2B}{6m}\langle r^2\rangle \\ &= -\frac{\mu_0 Ze^2H}{6m}\langle r^2\rangle \end{aligned} \tag{8.19}$$

and for N atoms per unit volume

$$\chi_{dia} = \frac{M}{H} = -\frac{N\mu_0 Ze^2}{6m}\langle r^2\rangle \tag{8.20}$$

This is Langevin's classical result, independent of temperature.

8.3.2 Quantum theory (background)

The field adds $\mathcal{H}'$ to the electron Hamiltonian:

$$\begin{aligned} \mathcal{H}' = &-\frac{ie\hbar}{2m}(\nabla\cdot\mathbf{A} + 2\mathbf{A}\cdot\nabla) \\ &+ \frac{e^2}{2m}A^2 \end{aligned} \tag{8.21}$$

For a uniform field,

$$\mathbf{A} = \frac{1}{2}\mathbf{B}\times\mathbf{r} \tag{8.22}$$

and with $\mathbf{B}$ along z:

$$A_x = -\frac{1}{2}yB, \quad A_y = \frac{1}{2}xB, \quad A_z = 0 \tag{8.23}$$

$$\begin{aligned}\mathcal{H}' = &-\frac{ie\hbar B}{2m}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) \\ &+ \frac{e^2B^2}{8m}(x^2+y^2)\end{aligned} \tag{8.24}$$

The angular momentum operator is

$$\begin{aligned}\mathbf{L} &= -i\hbar\,\mathbf{r}\times\nabla \\ &= -i\hbar\left[\hat{\mathbf{i}}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\right. \\ &\quad \left. +\hat{\mathbf{j}}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right) + \hat{\mathbf{k}}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)\right]\end{aligned} \tag{8.25}$$

The first term of (8.24) is proportional to L_z. The second term, summed over N atoms of Z electrons, is

$$\frac{Ne^2B^2}{8m}\sum_{i=1}^{Z}(x_i^2+y_i^2) = \frac{NZe^2B^2}{8m}\langle\rho^2\rangle \tag{8.26}$$

Comparing with $-\frac{1}{2\mu_0}\chi_{dia}B^2$:

$$\chi_{dia} = -\frac{N\mu_0 Ze^2}{4m}\langle\rho^2\rangle \tag{8.27}$$

$$\chi_{dia} = -\frac{N\mu_0 Ze^2}{6m}\langle r^2\rangle \tag{8.28}$$

This agrees with the classical result. $\langle r^2\rangle$ must be found from the electron charge distribution. Agreement is good for light elements and rare gases; for helium $\chi \approx -1.9\times10^{-6}$.

Worked Example 8.1

Ground-state hydrogen, $\psi = (\pi a_0^3)^{-1/2}e^{-r/a_0}$, $a_0 = 0.46$ Å, gives $\langle r^2\rangle = 3a_0^2$. With (8.28), $\chi_{dia} = -(N\mu_0 e^2/2m)a_0^2$, and with $N = 2.69\times10^{28}$ m^{-3} the book gives $\chi_{dia} = -1.01\times10^{-6}$ (SI).

Section 3: paramagnetism, Langevin and quantum theories (8.4, Examples 8.2 and 8.3)

8.4 Paramagnetism (structured notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 235–242 (§8.4 classical and quantum; Examples 8.2 and 8.3). Equations 8.29 to 8.51 and 8.83 are cited by their book numbers. Figures are redrawn.

Before you start

- Potential energy of a dipole, $U = -\boldsymbol{\mu}\cdot\mathbf{B}$ (Plus Two recall below).
- Maxwell–Boltzmann weighting $\exp(-E/kT)$.

> **PLUS TWO RECALL: A MAGNETIC DIPOLE IN A FIELD**
>
> **Recall.** $U = -\boldsymbol{\mu}\cdot\mathbf{B}$ for a dipole in a uniform field.
>
> **Re-derive.**
>
> 1. At angle θ, $U = -\mu B\cos\theta$.
> 2. Lowest energy at $\theta = 0$.
> 3. Thermal agitation spreads orientations, so only a partial alignment results.
>
> **Saved.** $U = -\mu B\cos\theta$ is the basis of Eq. (8.29).

8.4.1 Langevin's classical theory

For N independent dipoles of moment μ, the energy at angle θ is $E = -\mu B\cos\theta$, and the population in $d\omega$ follows the Maxwell–Boltzmann law:

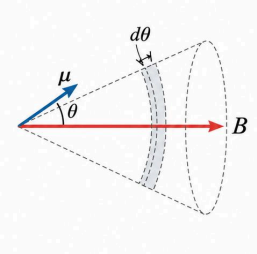


Fig. 8.1 (book): magnetic dipole with moment μ oriented at an angle θ to the applied magnetic field.

$$dn \propto \exp\left(\frac{\mu B\cos\theta}{kT}\right)d\omega = k\exp\left(\frac{\mu B\cos\theta}{kT}\right)2\pi\sin\theta\,d\theta \tag{8.29}$$

$$\begin{aligned} M &= N\mu\langle\cos\theta\rangle \\ &= N\frac{\mu\int_0^{\pi}\cos\theta\sin\theta\exp\left(\frac{\mu B\cos\theta}{kT}\right)d\theta}{\int_0^{\pi}\sin\theta\exp\left(\frac{\mu B\cos\theta}{kT}\right)d\theta} \end{aligned} \tag{8.30}$$

With $x = \mu B/kT$ and $y = \cos\theta$:

$$\begin{aligned} M &= N\mu\frac{\int_1^{-1} ye^{xy}\,dy}{\int_1^{-1} e^{xy}\,dy} \\ &= N\mu\left[\coth x - \frac{1}{x}\right] \\ &= N\mu\,L(x) \end{aligned} \tag{8.31}$$

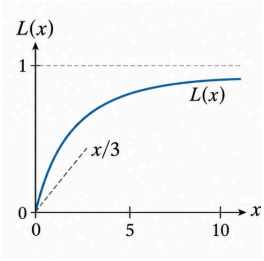


Fig. 8.2 (book): variation of L(x) with x.

For $x \ll 1$, $L(x) \cong x/3$:

$$L(x)|_{x=0} \cong \frac{x}{3} = \frac{\mu B}{3kT} \tag{8.32}$$

Hence

$$\chi_{para} = \frac{\mu_0 M}{B} = \frac{\mu_0 N\mu^2}{3kT} \tag{8.33}$$

$$\chi_{para} = \frac{C}{T} \tag{8.34}$$

with $C = \mu_0 N\mu^2/3k$ (Curie law, valid for $\mu B \ll kT$). For large x, saturation gives

$$M = N\mu = M_s \tag{8.35}$$

8.4.2 Quantum theory

The moment and its component are quantised:

$$\boldsymbol{\mu} = -g\mu_B\mathbf{J} \tag{8.36}$$

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \tag{8.37}$$

$$\mu_z = -g\mu_B m_J \tag{8.38}$$

$$E = m_J g\mu_B B \tag{8.39}$$

$$M = N\sum_{m_J=-J}^{m_J=+J}\frac{-m_J g\mu_B e^{-m_J g\mu_B B/kT}}{e^{-m_J g\mu_B B/kT}} \tag{8.40}$$

Weak fields, ordinary temperatures. Expanding the exponentials and using $\sum m_J = 0$, $\sum m_J^2 = J(J+1)(2J+1)/3$:

$$M = N\frac{g^2\mu_B^2 B}{3kT}J(J+1) \tag{8.41}$$

$$\chi_{para} = \frac{\mu_0 M}{B} = \frac{\mu_0 N\mu_B^2}{3kT}g^2 J(J+1) \tag{8.42}$$

$$\chi_{para} = \frac{\mu_0 N p_{eff}^2\mu_B^2}{3kT} \tag{8.43}$$

$$p_{eff} = g\sqrt{J(J+1)} \tag{8.44}$$

$$\mu^2 = p_{eff}^2\mu_B^2 \tag{8.45}$$

Strong fields, low temperatures.

$$M = N g J \mu_B B_J(x) \tag{8.46}$$

$$B_J(x) = \frac{2J+1}{2J}\coth\left(\frac{2J+1}{2J}x\right) - \frac{1}{2J}\coth\left(\frac{x}{2J}\right) \tag{8.47}$$

For $x \ll 1$, $B_J = x(J+1)/3J$, so

$$\begin{aligned} \chi_{para} &= \frac{\mu_0 M}{B} \\ &= \frac{\mu_0 N J(J+1) g^2 \mu_B^2}{3kT} \\ &= \frac{\mu_0 N p_{eff}^2 \mu_B^2}{3kT} \end{aligned} \tag{8.48}$$

For $x \gg 1$, $B_J \cong 1$ and

$$M = N g J \mu_B \tag{8.49}$$

As $J \to \infty$, $B_J \to L(x)$, so the classical result is recovered. When the crystal field quenches the orbital moment, $E = 2 m_s \mu_B B$ with $g = 2$ (8.50), and

$$E = 2 m_s \mu_B B \tag{8.50}$$

$$\chi = \frac{\mu_0 N}{3kT}\left(\frac{e\hbar}{m}\right)^2 S(S+1) \tag{8.51}$$

This gives the observed susceptibility of Mn^{3+} ($S = 2$), where (8.48) gives zero.

Worked Examples 8.2 and 8.3

Example 8.2 ($L = 0$, $S = \tfrac{1}{2}$, $J = \tfrac{1}{2}$, $g = 2$): with two levels,

$$M = g \mu_B J N \tanh\left(\frac{g\mu_B B}{2kT}\right) \tag{8.83}$$

For $N = 10^{28}$ m^{-3} and $B = 2.5$ T, $M = 519.1$ A/m at 300 K and $M = 3.0 \times 10^{4}$ A/m at 5 K; the level populations are given in the book.

Example 8.3 (Dy^{3+}, $S = 5/2$, $L = 5$, $J = 15/2$, $g = 1.33$): $\chi = 5.9 \times 10^{-4}$ at 300 K for one kilogram-mole.

Section 4: ferromagnetism: Weiss theory, exchange, domains and hysteresis (8.5, Example 8.4)

8.5 Ferromagnetism: Weiss theory, exchange, domains and hysteresis (structured notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 242–251 (§8.5, §8.5.1 to §8.5.3; Example 8.4). Equations 8.52 to 8.75 and 8.84 are cited by their book numbers. Figures are redrawn.

Before you start

- Brillouin function and saturated magnetisation (Eqs. 8.46, 8.47, 8.49).
- Molecular (Weiss) field idea: Beyond the Plus Two base.

> **PLUS TWO RECALL: HYSTERESIS OF MAGNETIC MATERIALS**
>
> **Recall.** Soft materials are easily magnetised and demagnetised; hard materials keep their magnetisation. Their behaviour is a loop in B against H.
>
> **Re-derive.**
>
> 1. Increasing H gives a rising magnetisation to saturation.
> 2. Reducing H to zero leaves the remanence M_r.
> 3. Reversing H to H_c removes it. The closed curve is the hysteresis loop.
>
> **Saved.** §8.5.3 and Fig. 8.6.

8.5.1 Weiss theory

Spontaneous magnetisation exists below T_C. Weiss proposed domains, each spontaneously magnetised, and an exchange field proportional to the magnetisation:

$$B_E = \lambda M \tag{8.52}$$

The effective field is

$$B_{eff} = B + B_E = B + \lambda M \tag{8.53}$$

Quantum form, with B_J as in (8.55):

$$M = NgJ\mu_B B_J(x) \tag{8.54}$$

$$B_J(x) = \frac{2J+1}{2J}\coth\left(\frac{2J+1}{2J}x\right) - \frac{1}{2J}\coth\left(\frac{x}{2J}\right) \tag{8.55}$$

$$x = \frac{gJ\mu_B B_{eff}}{kT} = \frac{gJ\mu_B}{kT}(B + \lambda M) \tag{8.56}$$

Spontaneous magnetisation ($B = 0$):

$$x = \frac{gJ\mu_B \lambda M}{kT} \tag{8.57}$$

$$M(T) = \frac{xkT}{\lambda g J \mu_B} \tag{8.58}$$

The saturation value is

$$M_s(0) = NgJ\mu_B \tag{8.59}$$

$$\frac{M(T)}{M_s(0)} = \frac{xkT}{\lambda N g^2 J^2 \mu_B^2} \tag{8.60}$$

$$\frac{M(T)}{M_s(0)} = B_J(x) \tag{8.61}$$

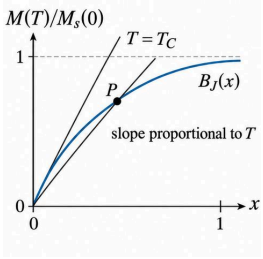


Fig. 8.3 (book): graphical solution of the simultaneous equations (8.60) and (8.61); the point of intersection P gives the spontaneous magnetisation $M_S(T)$.

The intersection of (8.60) and (8.61) gives $M_s(T)$ for $T < T_C$. It vanishes at T_C, a second-order transition. Spontaneous magnetisation versus temperature is shown in Fig. 8.4.

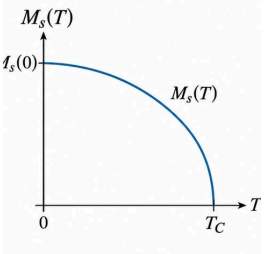


Fig. 8.4 (book): spontaneous magnetisation against temperature for $T < T_C$.

The paramagnetic region

For $T > T_C$, $x \ll 1$ and $B_J \cong (J+1)x/3J$. Then

$$x = \frac{gJ\mu_B(B+\lambda M)}{kT} \tag{8.62}$$

$$M = \frac{Ng^2\mu_B^2 J(J+1)}{3kT}(B+\lambda M) \tag{8.63}$$

$$\chi = \frac{\mu_0 M}{B} = \frac{\mu_0 T_C/\lambda}{T-T_C} = \frac{C}{T-T_C} \tag{8.64}$$

$$C = \frac{\mu_0 T_C}{\lambda} \tag{8.65}$$

$$T_C = \frac{\lambda N g^2 \mu_B^2 J(J+1)}{3k} \tag{8.66}$$

(8.64) is the Curie–Weiss law.

8.5.2 Origin of the Weiss field

Dipole–dipole fields are about 10^3 G, far below the Weiss field of about 10^7 G for iron. Heisenberg attributed it to exchange, a non-magnetic electrostatic effect that depends on wave-function overlap and spin orientation:

$$U_{ij} = -2J_e \mathbf{S}_i \cdot \mathbf{S}_j \tag{8.67}$$

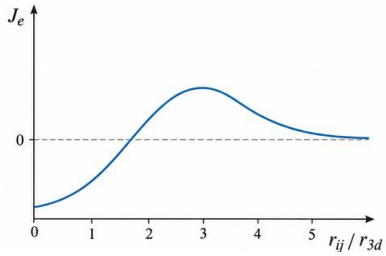


Fig. 8.5 (book): exchange integral J_e against the ratio of interatomic separation to the radius of the 3d orbit.

Stoner's mean-field treatment of z neighbours gives

$$U_i = -2J_e \sum_j \mathbf{S}_i \cdot \mathbf{S}_j \tag{8.68}$$

$$U_i = -2zJ_e\left(S_{xi}\langle S_{xj}\rangle + S_{yi}\langle S_{yj}\rangle + S_{zi}\langle S_{zj}\rangle\right) \tag{8.69}$$

With $\langle S_{xj}\rangle = \langle S_{yj}\rangle = 0$ and

$$\langle S_{zj}\rangle = \frac{M}{g\mu_B N} \tag{8.70}$$

the exchange energy becomes

$$U_i = -\frac{2zJ_e S_{zi} M}{gN\mu_B} \tag{8.71}$$

$$U_i = -gS_{zi}\mu_B B_E \tag{8.72}$$

$$B_E = \frac{2zJ_e}{Ng^2\mu_B^2} M \tag{8.73}$$

$$\lambda = \frac{2zJ_e}{Ng^2\mu_B^2} \tag{8.74}$$

so $B_E = \lambda M$ (8.52).

8.5.3 Domains and hysteresis

Domains have random orientations, so there is no net magnetisation. Field growth of domains (Fig. 8.7b) and their rotation (Fig. 8.7c) produce the magnetisation.

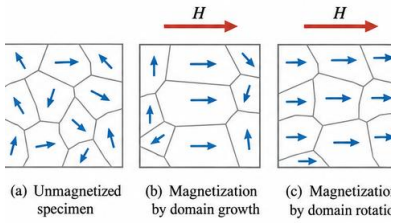


Fig. 8.7 (book): the two fundamental processes of magnetisation: (a) unmagnetised specimen; (b) magnetisation by domain growth; (c) magnetisation by domain rotation.

Neel's energy minimisation (magnetic field, anisotropy, domain wall and magnetostrictive energy) fixes the domain size. For $T < T_C$ the hysteresis boundary is traced by the intersections of (8.61) with

$$M = \left(\frac{kT}{\mu_B g\lambda J}\right)x - \frac{1}{\lambda}B \tag{8.75}$$

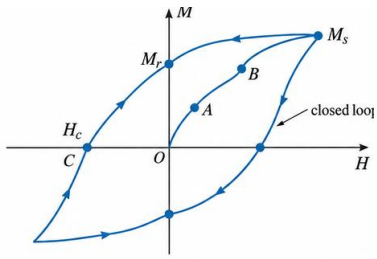


Fig. 8.6 (book): typical hysteresis curve for a ferromagnetic material, indicating the predominant magnetisation processes in different regions.

OA (reversible), AB (large irreversible domain growth), BC (rotation to M_s), then remanence M_r and coercivity H_c close the loop CEFKC.

Worked Example 8.4

$J = 3/2$, $g = 2$, $T_C = 125$ K:
$B_E = 3kT_C/[g\mu_B(J+1)] = 111.6$ T. Above T_C,

$$M = \frac{Ng^2\mu_B^2 J(J+1)B}{3k(T - T_C)} \tag{8.84}$$

giving $M/M_s(0) = 6.4 \times 10^{-6}$ for $B = 1$ mT at 300 K.

Section 5: antiferromagnetism and ferrimagnetism (8.6–8.7)

8.6–8.7 Antiferromagnetism and ferrimagnetism (structured notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 251–255 (§8.6, §8.7). Equations 8.76 to 8.82 are cited by their book numbers. Figures are redrawn. Fig. 8.9 has no caption in the book. Item 19 is qualitative; the molecular-field algebra is given as the book's picture.

Before you start

- Curie–Weiss law (Eq. 8.64) and paramagnetic law (Eq. 8.33).
- Sign of the exchange integral (§8.5.2).

8.6 Antiferromagnetism

Spins of neighbouring atoms are antiparallel, forming two interpenetrating sublattices A and B with opposite magnetisation (Fig. 8.8). The net moment is zero. MnO was the first observed example. A field gives a small magnetisation that grows with temperature up to the Néel temperature T_N, above which the material is paramagnetic. The susceptibility has a weak cusp at T_N (Fig. 8.9).

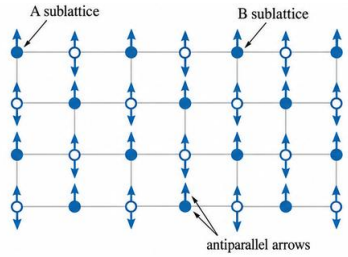


Fig. 8.8 (book): neighbouring antiparallel spin moments of the A and B sublattices producing antiferromagnetism.

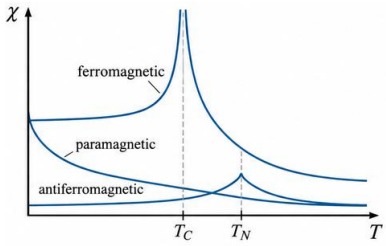


Fig. 8.9 (book): the book prints the number only; no caption is printed.

Molecular-field treatment of Néel

With α (AA, BB) and β (AB) as the Weiss field constants, the internal fields are

$$\begin{aligned} B_{mA} &= -\alpha M_A - \beta M_B \\ B_{mB} &= -\beta M_A - \alpha M_B \end{aligned} \tag{8.76}$$

and with the applied field B:

$$\begin{aligned} B_A &= B - \alpha M_A - \beta M_B \\ B_B &= B - \beta M_A - \alpha M_B \end{aligned} \tag{8.77}$$

$T > T_N$. Each sublattice follows the paramagnetic law:

$$M_A = \frac{N\mu^2}{3kT} B_A \qquad M_B = \frac{N\mu^2}{3kT} B_B \tag{8.78}$$

with $\mu^2 = \mu_B^2 g^2 J(J+1)$. Summing with $N_A = N_B = N$:

$$\begin{aligned} M &= M_A + M_B \\ &= \frac{N\mu^2}{3kT}(B_A + B_B) \\ &= \frac{N\mu^2}{3kT}[2B - (\alpha+\beta)(M_A + M_B)] \\ &= \frac{N\mu^2}{3kT}[2B - (\alpha+\beta)M] \end{aligned} \tag{8.79}$$

$$\begin{aligned} \chi &= \frac{\mu_0 M}{B} \\ &= \frac{\dfrac{2\mu_0 N\mu^2}{3k}}{T + \dfrac{N\mu^2(\alpha+\beta)}{3k}} \\ &= \frac{C}{T+\theta} \end{aligned} \tag{8.80}$$

$$C = \frac{2\mu_0 N\mu^2}{3k}$$

$$\theta = \frac{N\mu^2(\alpha+\beta)}{3k} \tag{8.81}$$

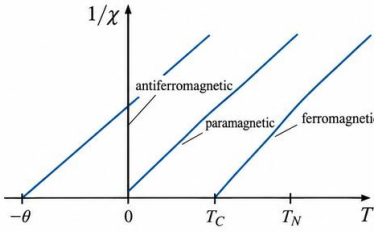


Fig. 8.10 (book): plot of 1/χ against T for para-, ferro- and antiferromagnetic substances above their critical temperatures.

$$\frac{T_N}{\theta} = \frac{\beta - \alpha}{\beta + \alpha} \tag{8.82}$$

For $\alpha = 0$, $T_N = \theta$; $T_N < \theta$ for positive α; T_N rises with β.

$T < T_N$. $\chi_\perp$ is almost constant; $\chi_\parallel$ falls to zero as $T \to 0$; both equal χ at T_N (Fig. 8.11).

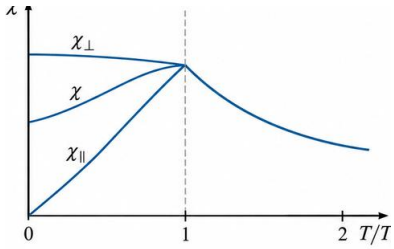


Fig. 8.11 (book): plot of χ against T/T_N for an antiferromagnetic single crystal.

8.7 Ferrimagnetism

The sublattice magnetisations differ in size, so there is a net moment. Ferrites such as magnetite, $\mathbf{Fe_3O_4}$, have one $\mathbf{Fe^{3+}}$ on tetrahedral A sites and one $\mathbf{Fe^{3+}}$ and one $\mathbf{Fe^{2+}}$ on octahedral B sites. The net moment of a formula unit is that of one $\mathbf{Fe^{2+}}$ ion (Fig. 8.12), and a spinel cell has eight formula units.

Ferrimagnets show spontaneous magnetisation and hysteresis like ferromagnets. Néel's theory explains the antiparallel alignment through the negative AB interaction; above T_C the sublattices obey the Curie–Weiss law, below it Eq. (8.46) applies. The detailed theory is outside the book's scope.

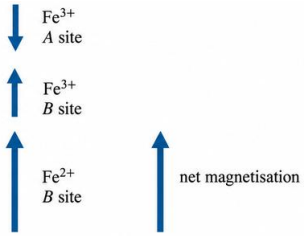


Fig. 8.12 (book): magnetisations of the individual sites and the net magnetisation in one formula unit of Fe_3O_4.

Section 6: polarisation, local field and dielectric constant (9.1–9.3)

9.1–9.3 Polarisation, local field and dielectric constant (structured notes)

Source: book-sources/bs-7-dielectrics.html, Puri and Babbar pp. 265–267. Equations 9.1 to 9.13 are cited by their book numbers. Figure is redrawn.

Before you start

- Capacitance with a dielectric, $C = KC_0$ (Plus Two recall below).
- Electric dipole moment $p = qd$.

PLUS TWO RECALL: A DIELECTRIC IN A CAPACITOR

Recall. A dielectric of constant K between capacitor plates gives $C = KC_0$.

Re-derive.

1. Polarisation creates bound charges that oppose the free-charge field.
2. The resultant field is therefore smaller than the applied field.
3. For fixed charge, C rises by the factor K.

Saved. Eq. (9.1) and Eq. (9.5).

9.1 Polarisation and susceptibility

The polarisation field opposes the applied field, so

$$\mathbf{E} = \mathbf{E}_o + \mathbf{E}_p \tag{9.1}$$

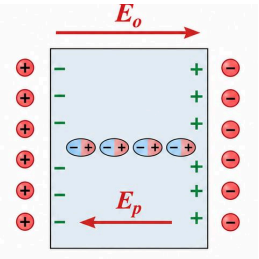


Fig. 9.1 (book): a dielectric slab in an external field E_o produced by fixed charges outside the slab; the internal polarisation field E_p is due to fictitious bound charges at the surface.

$$\mathbf{P} = \varepsilon_o \chi_e \mathbf{E} \tag{9.2}$$

where χ_e is the electric susceptibility.

9.2 The local field

The local field at an atom with cubic symmetry is given by the Lorentz relation:

$$\begin{aligned} \mathbf{E}_{loc} &= \mathbf{E}_o + \mathbf{E}_p + \frac{\mathbf{P}}{3\varepsilon_o} \\ &= \mathbf{E} + \frac{\mathbf{P}}{3\varepsilon_o} \end{aligned} \tag{9.3}$$

The extra term is the field of the other atoms' polarisation. Its form depends on specimen shape.

9.3 Dielectric constant and polarisability

$$\begin{aligned} \mathbf{D} &= \varepsilon_o \varepsilon_r \mathbf{E} \\ &= \varepsilon_o \mathbf{E} + \mathbf{P} \end{aligned} \tag{9.4}$$

$$\begin{aligned} \varepsilon_r &= \frac{\varepsilon_o \mathbf{E} + \mathbf{P}}{\varepsilon_o \mathbf{E}} \\ &= 1 + \chi_e \end{aligned} \tag{9.5}$$

The polarisability of an atom is

$$p = \alpha E_{loc} \tag{9.6}$$

Polarisability is atomic; the dielectric constant is macroscopic. For N identical atoms per unit volume:

$$P = Np = N\alpha E_{loc} \tag{9.7}$$

$$P = \sum_j N_j \alpha_j E_{loc}(j) \tag{9.8}$$

With cubic symmetry, (9.3) and (9.7) give

$$P = N\alpha \left(E + \frac{P}{3\varepsilon_o} \right) \tag{9.9}$$

which yields

$$\begin{aligned} \chi_e &= \frac{P}{\varepsilon_o E} \\ &= \frac{N\alpha/\varepsilon_o}{1 - \dfrac{N\alpha}{3\varepsilon_o}} \end{aligned} \tag{9.10}$$

$$\varepsilon_r = \frac{1 + \dfrac{2N\alpha}{3\varepsilon_o}}{1 - \dfrac{N\alpha}{3\varepsilon_o}} \tag{9.11}$$

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_o} \tag{9.12}$$

This is the Clausius–Mossotti relation. In general form:

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{1}{3\varepsilon_o}\sum_j N_j\alpha_j \qquad (9.13)$$

Section 7: sources of polarisability and frequency dependence (9.4–9.5, Example 9.1)

9.4–9.5 Sources of polarisability and its frequency dependence (structured notes)

Source: book-sources/bs-7-dielectrics.html, Puri and Babbar pp. 267–272 (§9.4, §9.5; Example 9.1). Equations 9.14 to 9.32 and 9.40 to 9.42 are cited by their book numbers. Figures are redrawn.

Before you start

- Clausius–Mossotti relation (Eq. 9.12).
- Potential energy $U = -\mathbf{p}\cdot\mathbf{E}$ (Plus Two recall below).

> **PLUS TWO RECALL: A DIPOLE IN AN ELECTRIC FIELD**
>
> **Recall.** $U = -\mathbf{p}\cdot\mathbf{E}$ for a dipole in a uniform field.
>
> **Re-derive.**
>
> 1. At angle θ, $U = -pE\cos\theta$.
> 2. Thermal agitation gives a partial alignment that depends on pE/kT.
> 3. This gives the orientational polarisability $p^2/3kT$.
>
> **Saved.** Eqs. (9.24) and (9.31).

9.4 Sources of polarisability

Electronic, ionic and dipolar (orientational) polarisability make up the net value. Their relative weight depends on the material and on the frequency.

9.4.1 Electronic polarisability

Electronic polarisation is the displacement of the electron cloud relative to the nucleus (Fig. 9.2). At optical frequencies the dielectric constant is mainly electronic. With $\varepsilon_r = n^2$:

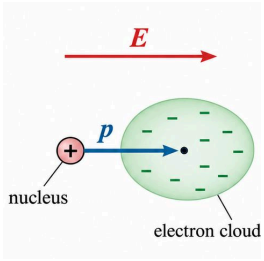


Fig. 9.2 (book): electronic polarisation in the presence of an external field.

$$\frac{n^2 - 1}{n^2 + 2} = \frac{1}{3\varepsilon_o}\sum_j N_j\alpha_j \quad \text{(electronic)} \qquad (9.14)$$

Classical resonance and static polarisability:

$$\omega_o = \sqrt{\beta/m} \qquad (9.15)$$

$$-eE_{loc} = \beta x = m\omega_o^2 x \qquad (9.16)$$

$$\alpha_e = \frac{p}{E_{loc}} = -\frac{ex}{E_{loc}} = \frac{e^2}{m\omega_o^2} \qquad (9.17)$$

$$m\frac{d^2x}{dt^2} + m\omega_o^2 x = -eE_{loc}\sin\omega t \qquad (9.18)$$

$$m(-\omega^2 + \omega_o^2)x_o = -eE_{loc} \qquad (9.19)$$

$$\alpha_e = \frac{p_o}{E_{loc}} = \frac{e^2/m}{\omega_o^2 - \omega^2} \qquad (9.20)$$

$$\alpha_e = \frac{e^2}{m}\sum_j \frac{f_{ij}}{\omega_{ij}^2 - \omega^2} \tag{9.21}$$

9.4.2 Ionic polarisability

$$\beta \Delta x \cong eE \tag{9.22}$$

$$\alpha_i = \frac{p}{E} = \frac{e\Delta x}{E} \cong \frac{e^2}{\beta} \tag{9.23}$$

For $\beta \simeq 20$ N/m, $\alpha_i \simeq 10^{-39}$ F m^2. NaCl: $\varepsilon_r \simeq 5.6$ at low frequency, 2.25 at optical frequencies.

9.4.3 Dipolar polarisability

A polar molecule has a permanent dipole. The energy at angle θ (Fig. 9.3) is

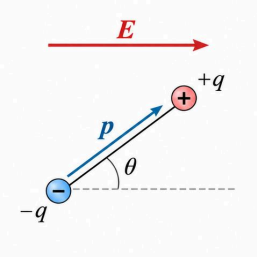


Fig. 9.3 (book): a molecule of dipole moment p placed in an electric field E.

$$U = -\mathbf{p}\cdot\mathbf{E} = -pE\cos\theta \tag{9.24}$$

$$\exp\left(-\frac{U}{kT}\right) d\Omega = \exp\left(\frac{pE\cos\theta}{kT}\right) 2\pi \sin\theta\, d\theta \tag{9.25}$$

$$2\pi p \exp\left(\frac{pE\cos\theta}{kT}\right)\cos\theta\sin\theta\, d\theta \tag{9.26}$$

$$\bar{p} = \frac{P}{N} = \frac{p\int_0^\pi \exp\left(\frac{pE\cos\theta}{kT}\right)\cos\theta\sin\theta\, d\theta}{\int_0^\pi \exp\left(\frac{pE\cos\theta}{kT}\right)\sin\theta\, d\theta} \tag{9.27}$$

$$\frac{\bar{p}}{p} = \coth x - \frac{1}{x} = L(x) \tag{9.28}$$

$$L(x) \simeq \frac{x}{3} = \frac{pE}{3kT} \tag{9.29}$$

$$P = N\bar{p} = \frac{Np^2E}{3kT} \tag{9.30}$$

$$\alpha_d = \frac{\bar{p}}{E} = \frac{p^2}{3kT} \tag{9.31}$$

9.5 Frequency dependence of the total polarisability

$$\alpha = \alpha_e + \alpha_i + \alpha_d \tag{9.32}$$

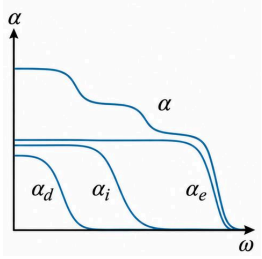


Fig. 9.4 (book): frequency dependence of the various contributions to polarisability.

The total falls with frequency. Relaxation times are longest for dipolar, then ionic, then electronic processes, so the dipolar part disappears first.

Worked Example 9.1

NaCl with $n = 1.5$, $\varepsilon_r = 5.6$. The static and optical Clausius–Mossotti relations are

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N(\alpha_e + \alpha_i)}{3\varepsilon_0} \tag{9.40}$$

$$\frac{n^2 - 1}{n^2 + 2} = \frac{N\alpha_e}{3\varepsilon_0} \tag{9.41}$$

Dividing (9.41) by (9.40):

$$\left(\frac{n^2-1}{n^2+2}\right)\left(\frac{\varepsilon_r+2}{\varepsilon_r-1}\right) = \frac{\alpha_e}{\alpha_e+\alpha_i} \tag{9.42}$$

The ionic share is $[1 - (9.42)] \times 100 = 51.4\%$.

9.6–9.7 Ferroelectricity and piezoelectricity (structured notes)

Source: book-sources/bs-7-dielectrics.html, Puri and Babbar pp. 272–276 (§9.6, §9.7). Equations 9.33 to 9.39 are cited by their book numbers. Figures are redrawn. Item 22 is qualitative only.

Before you start

- Clausius–Mossotti relation (Eqs. 9.12, 9.13).
- Ferromagnetism as the analogy (§8.5).

Beyond the Plus Two base (pre item P12).

9.6 Ferroelectricity

Ferroelectricity is spontaneous polarisation, the electrical analogue of spontaneous magnetisation. Centres of positive and negative charge do not coincide in the absence of a field. The polarisation against field forms a hysteresis loop. Above the Curie point T_C the crystal is paraelectric and the dielectric constant falls sharply.

Examples are $\mathbf{BaTiO_3}$ and Rochelle salt. Order–disorder ferroelectrics (for example $\mathbf{KH_2PO_4}$, $\mathbf{RbH_2PO_4}$) transform by ionic ordering, and displacive ones ($\mathbf{BaTiO_3}$, $\mathbf{KNbO_3}$, $\mathbf{LiTaO_3}$) by the displacement of one sublattice.

In $\mathbf{BaTiO_3}$ above about 380 K the cell is cubic, with $\mathbf{Ba^{2+}}$ at the corners, $\mathbf{O^{2-}}$ at the face centres and $\mathbf{Ti^{4+}}$ at the body centre (Fig. 9.5). Below T_C the Ti and Ba ions move up and the O ions move down, the structure becomes tetragonal, and a spontaneous polarisation appears.

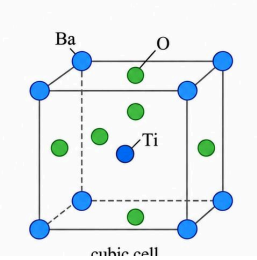


Fig. 9.5 (book): structure of $BaTiO_3$ for $T > T_C$.

$$\varepsilon_r = \frac{1 + \frac{2}{3\varepsilon_o}\sum_j N_j\alpha_j}{1 - \frac{1}{3\varepsilon_o}\sum_j N_j\alpha_j} \tag{9.33}$$

The dielectric constant diverges when

$$\sum_j N_j\alpha_j = 3\varepsilon_o \tag{9.34}$$

This is the polarisation catastrophe: the local field grows faster than the restoring force, and higher-order forces limit the displacement. Writing $\frac{1}{3\varepsilon_o}\sum_j N_j\alpha_j = 1 - 3s$ with $s \ll 1$:

$$\frac{1}{3\varepsilon_o}\sum_j N_j\alpha_j = 1 - 3s \tag{9.35}$$

$$\varepsilon_r = \frac{1}{s} - 2 \cong \frac{1}{s} \tag{9.36}$$

$$s \cong \frac{T - T_C}{\xi} \tag{9.37}$$

$$\varepsilon_r \cong \frac{\xi}{T - T_C} \tag{9.38}$$

This matches the paraelectric state qualitatively.

9.7 Piezoelectricity

A stress induces polarisation in a piezoelectric crystal, and an electric field strains it (quartz, Rochelle salt, tourmaline). In one-dimensional notation:

$$\begin{aligned} P &= Zd + \varepsilon_o E\chi \\ e &= Zs + Ed \end{aligned} \tag{9.39}$$

Strains are very small: in quartz about $\mathbf{10^{-8}}$ for $\mathbf{10^4}$ V/m. Piezoelectricity needs the absence of a centre of inversion (Fig. 9.6). All ferroelectrics are piezoelectric; not all piezoelectrics are ferroelectric ($\mathbf{BaTiO_3}$ and quartz are the examples). The devices are transducers: gramophone pick-ups, microphones, strain gauges and ultrasonic generators.

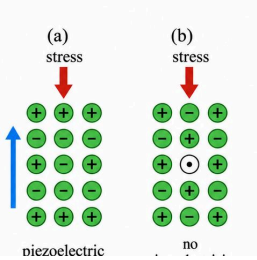


Fig. 9.6 (book): effect of crystal symmetry on piezoelectricity: (a) quartz, with no centre of inversion, shows piezoelectricity; (b) a crystal with a centre of inversion shows none.